

Yes, you can transform the site into an interactive 3D motion website while preserving the exact layout, geometry, and design architecture.

Because the existing project is built on React, Vite, and TypeScript with a strict component architecture, adding 3D motion does not require rebuilding from scratch. Instead, you can layer a WebGL/3D render engine directly into your existing React setup while strictly observing the established brand governance and performance budgets.

Key Technical Strategy for a 3D Motion Portfolio

1. Tech Stack Integration: React Three Fiber (R3F)

Since your project is React-based, the ideal choice is **React Three Fiber (@react-three/fiber)** alongside **@react-three/drei**.

- **Why R3F:** It hooks seamlessly into React state, allowing you to pass props like active discipline ('video' | 'motion' | 'voice') or hover states directly into 3D meshes without dropping out of your existing component tree.
- **Alternative: Rive 3D / Spline** can be used for interactive canvas embeds, though Three.js/R3F offers fine-grained control over custom shaders, glassmorphism, and hardware performance scaling.

2. Vector-True 3D Mesh Generation

To ensure zero visual drift from the certified geometry, do not manually model the logo in external software without referencing the source.

- Use Three.js's `SVGLoader` alongside `ExtrudeGeometry` to import `public/hydra-mark.svg` programmatically.
- Extruding the exact path coordinates creates a clean 3D bevel while retaining the 0.0° joint tangency and 57.5° head flare of the V4 master.

3. Aligning 3D Visuals with Brand Identity

To match the **Bio-Organic Dark Editorial** aesthetic, configure the 3D scene using your established design tokens:

- **Materials:** Apply a `MeshPhysicalMaterial` with low roughness, high metalness, transmission/glass effects, and an emissive glow map mapped to #10b981 (Mythic Emerald).
- **Atmosphere:** Set the 3D scene background to #060c09 (Abyssal Slate) with point lights placed behind the extruded mark using #34d399 (Soft Mint) for volumetric halo lighting.
- **Camera Interaction:** Implement subtle mouse/pointer parallax tracking (using `@react-three/drei`'s `PresentationControls` or `Float`) so the 3D mark responds fluidly as the cursor moves across the viewport.

4. Strict Compliance with Motion Governance

- **No Continuous Spinning:** Brand governance strictly bans continuous 360° spinning animations (`.hydra-mark-spin`), as continuous rotation triggers fan or rotor blade misinterpretations.
- **Approved 3D Motion:** Stick to organic depth-tilts (≤ 15°), z-axis camera pushes on

scroll, subtle floating pulses, and geometry draw/extrusion reveals on initial page load.

5. Linking to the Adaptive Performance Tier System

3D WebGL scenes can be resource-intensive on mobile or lower-spec devices. You can wire the Three.js `<Canvas>` directly into your existing `useDeviceTier` hook (`html[data-quality]`):

- **High Tier:** Render full 3D extruded mesh, post-processing bloom, SSAO depth-of-field, and fluid camera parallax.
- **Medium Tier:** Render the 3D mesh with basic PBR lighting, reduced shadow maps, and disable heavy post-processing shaders.
- **Low Tier:** Automatically unmount the 3D canvas and fall back to the optimized 2D vector `HydraLogo.tsx` component.

Step-by-Step AI Workflow (OpenCode & AI Studio)

When using **OpenCode with Big Pickle / Claude / Gemini AI Studio**, prompt incrementally to prevent code regression:

1. **Phase 1 — Dependency & Shell Setup:**
 - Ask OpenCode to install `three`, `@types/three`, `@react-three/fiber`, and `@react-three/drei`.
 - Request a isolated `HydraCanvas3D.tsx` container component placed inside `src/components/` that sets up a basic `<Canvas>` with ambient lighting, camera controls, and responsive sizing.
2. **Phase 2 — Extruding the Certified Geometry:**
 - Feed your `HEAD_PATH` and `CORE_PATH` strings from `HydraLogo.tsx` into Google AI Studio / OpenCode.
 - Prompt: *"Write a React Three Fiber component using `SVGLoader` or `ShapePath` to parse these SVG path strings and extrude them into 3D meshes with rounded bevels."*
3. **Phase 3 — Materials & Shaders:**
 - Prompt: *"Apply `MeshPhysicalMaterial` using emissive `#10b981` (Mythic Emerald) accents on an `#060c09` dark canvas. Add a subtle floating animation using Drei's `<Float>` wrapper with floating range restricted to prevent tumbling."*
4. **Phase 4 — Scroll Integration & Fallbacks:**
 - Wire camera animation to scroll offset using `Framer Motion` or `Lenis`.
 - Ensure that `useDeviceTier` conditionally renders `HydraCanvas3D` on high/medium tiers and falls back to `HydraLogo.tsx` on low hardware tiers.